

# *Take a Sip of Data*

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# Exploratory Question

*Do red and white wines have different drivers of quality?*



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# Project Objective

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1. Predict **wine quality** using chemical measurements
  2. Evaluate **ensemble machine learning models**
  3. Compare key **predictors of quality** for red vs white wines
- 

## Models Used:

1. Random Forest Regressor
2. Gradient Boosting Regressor
3. Bagging Regressor



# Wine Quality Dataset



Red Wine

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1,599 samples



White Wine

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4,898 samples

**Total: 6,497 wines**

Each Sample Includes:

- 11 chemical attributes
- Quality score (0—10)



# Key Dataset Features

Predictor variables:

|               |                  |
|---------------|------------------|
| Fixed acidity | Volatile acidity |
| Citric acid   | Residual sugar   |
| Chlorides     | Sulphates        |
| Alcohol       | pH               |



# Summary Statistics

Red

|       | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides   | free sulfur dioxide | total sulfur dioxide | density     | pH          | sulphates   | alcohol     |
|-------|---------------|------------------|-------------|----------------|-------------|---------------------|----------------------|-------------|-------------|-------------|-------------|
| count | 1599.000000   | 1599.000000      | 1599.000000 | 1599.000000    | 1599.000000 | 1599.000000         | 1599.000000          | 1599.000000 | 1599.000000 | 1599.000000 | 1599.000000 |
| mean  | 8.319637      | 0.527821         | 0.270976    | 2.538806       | 0.087467    | 15.874922           | 46.467792            | 0.996747    | 3.311113    | 0.658149    | 10.422983   |
| std   | 1.741096      | 0.179060         | 0.194801    | 1.409928       | 0.047065    | 10.460157           | 32.895324            | 0.001887    | 0.154386    | 0.169507    | 1.065668    |
| min   | 4.600000      | 0.120000         | 0.000000    | 0.900000       | 0.012000    | 1.000000            | 6.000000             | 0.990070    | 2.740000    | 0.330000    | 8.400000    |
| 25%   | 7.100000      | 0.390000         | 0.090000    | 1.900000       | 0.070000    | 7.000000            | 22.000000            | 0.995600    | 3.210000    | 0.550000    | 9.500000    |
| 50%   | 7.900000      | 0.520000         | 0.260000    | 2.200000       | 0.079000    | 14.000000           | 38.000000            | 0.996750    | 3.310000    | 0.620000    | 10.200000   |
| 75%   | 9.200000      | 0.640000         | 0.420000    | 2.600000       | 0.090000    | 21.000000           | 62.000000            | 0.997835    | 3.400000    | 0.730000    | 11.100000   |
| max   | 15.900000     | 1.580000         | 1.000000    | 15.500000      | 0.611000    | 72.000000           | 289.000000           | 1.003690    | 4.010000    | 2.000000    | 14.900000   |

White

|       | fixed acidity | volatile acidity | citric acid | residual sugar | chlorides   | free sulfur dioxide | total sulfur dioxide | density     | pH          | sulphates   | alcohol     |
|-------|---------------|------------------|-------------|----------------|-------------|---------------------|----------------------|-------------|-------------|-------------|-------------|
| count | 4898.000000   | 4898.000000      | 4898.000000 | 4898.000000    | 4898.000000 | 4898.000000         | 4898.000000          | 4898.000000 | 4898.000000 | 4898.000000 | 4898.000000 |
| mean  | 6.854788      | 0.278241         | 0.334192    | 6.391415       | 0.045772    | 35.308085           | 138.360657           | 0.994027    | 3.188267    | 0.489847    | 10.514267   |
| std   | 0.843868      | 0.100795         | 0.121020    | 5.072058       | 0.021848    | 17.007137           | 42.498065            | 0.002991    | 0.151001    | 0.114126    | 1.230621    |
| min   | 3.800000      | 0.080000         | 0.000000    | 0.600000       | 0.009000    | 2.000000            | 9.000000             | 0.987110    | 2.720000    | 0.220000    | 8.000000    |
| 25%   | 6.300000      | 0.210000         | 0.270000    | 1.700000       | 0.036000    | 23.000000           | 108.000000           | 0.991723    | 3.090000    | 0.410000    | 9.500000    |
| 50%   | 6.800000      | 0.260000         | 0.320000    | 5.200000       | 0.043000    | 34.000000           | 134.000000           | 0.993740    | 3.180000    | 0.470000    | 10.400000   |
| 75%   | 7.300000      | 0.320000         | 0.390000    | 9.900000       | 0.050000    | 46.000000           | 167.000000           | 0.996100    | 3.280000    | 0.550000    | 11.400000   |
| max   | 14.200000     | 1.100000         | 1.660000    | 65.800000      | 0.346000    | 289.000000          | 440.000000           | 1.038980    | 3.820000    | 1.080000    | 14.200000   |

# Explorator y Data Analysis

*EDCA*



# Data Exploration

- No missing values in either dataset
- Quality scores mostly between 3–8
- Alcohol and acidity variables show strong relationships with quality



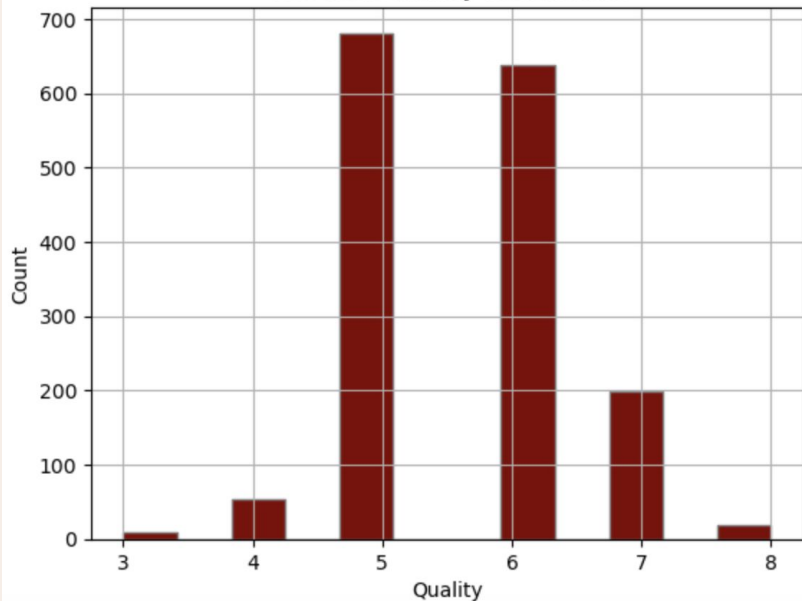
*Whoa!*



# EDA: Distribution

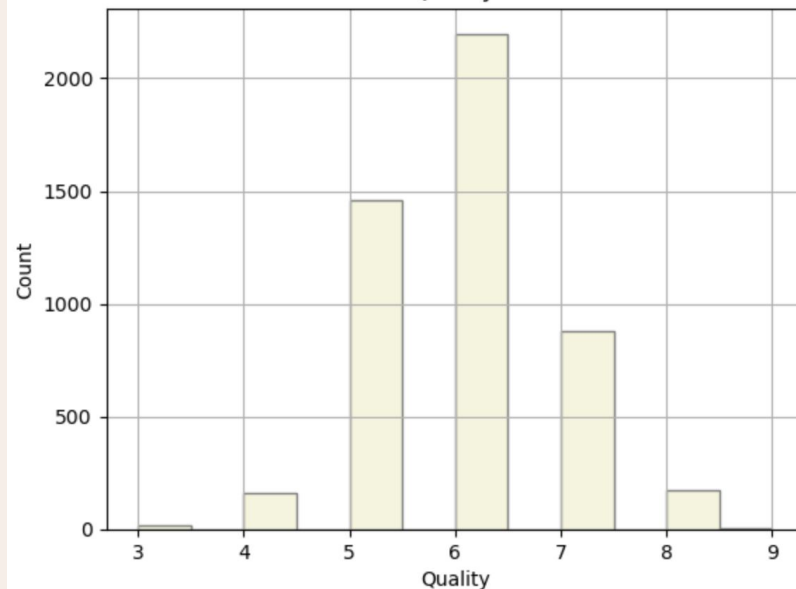
## Red Wine

Red Wine Quality Distribution



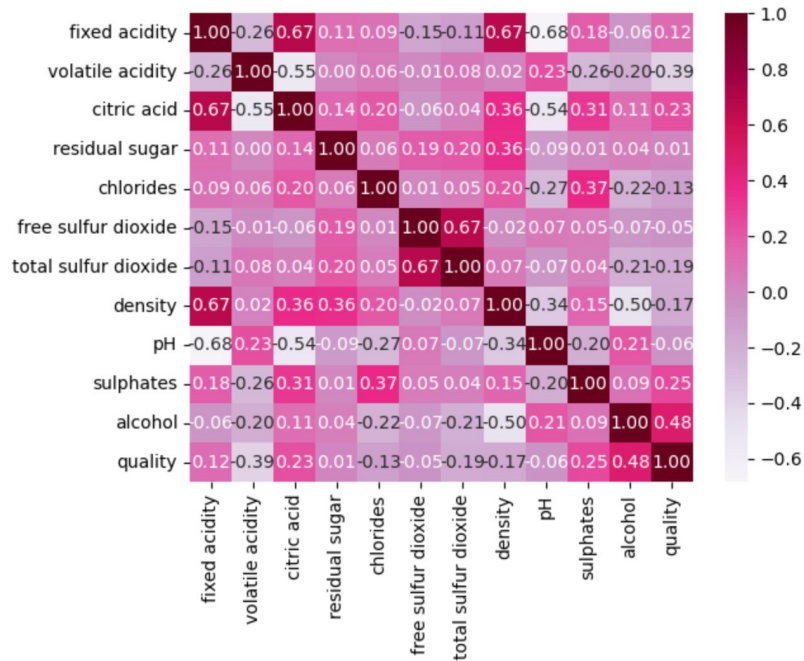
## White Wine

White Wine Quality Distribution

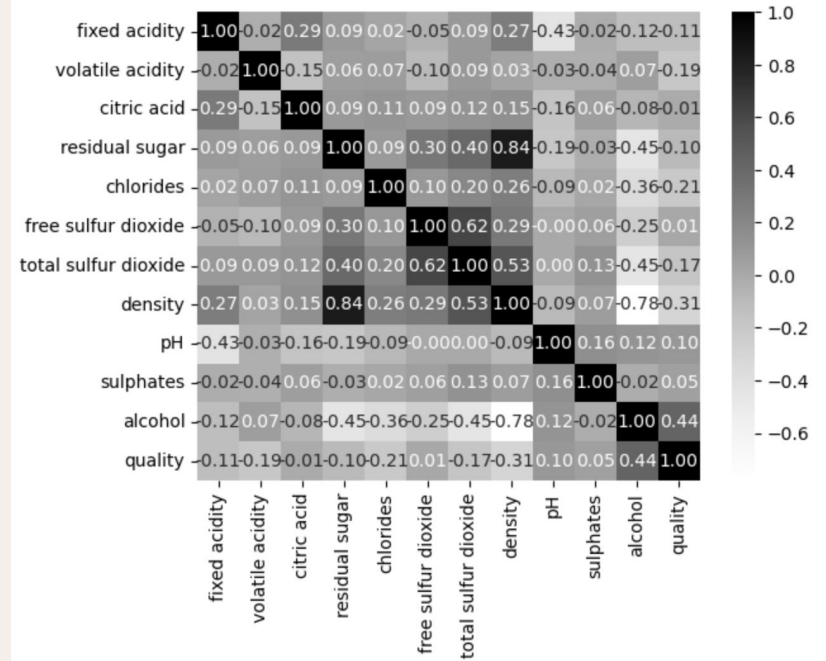


# EDA: Correlation Matrix

## Red Wine



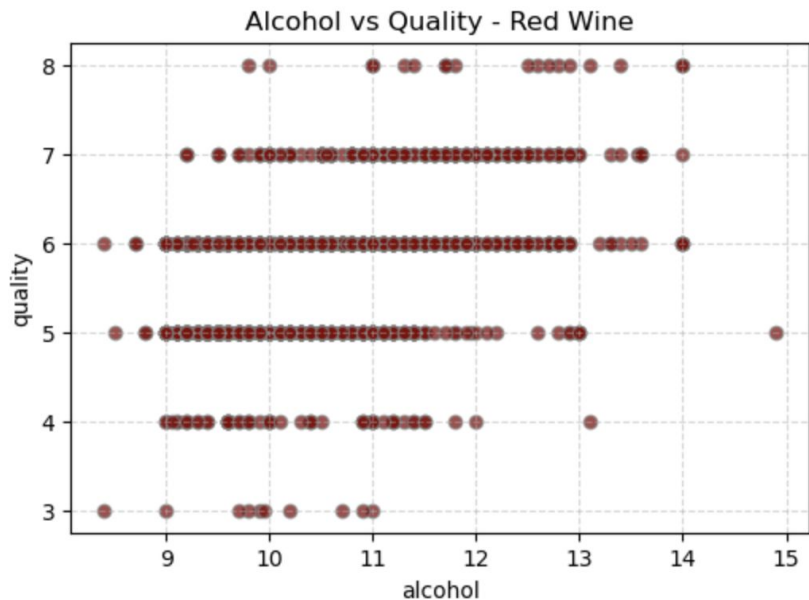
## White Wine



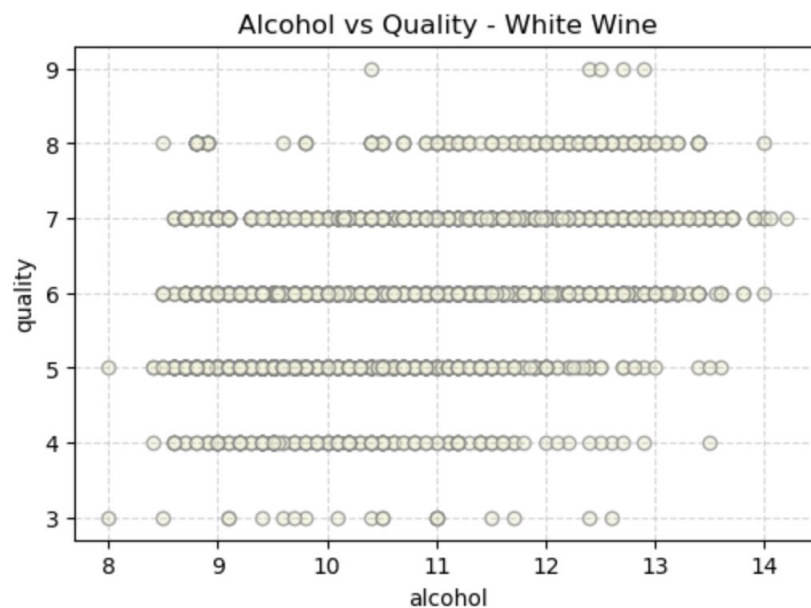


# EDA: Alcohol vs Quality Scatterplot

## Red Wine

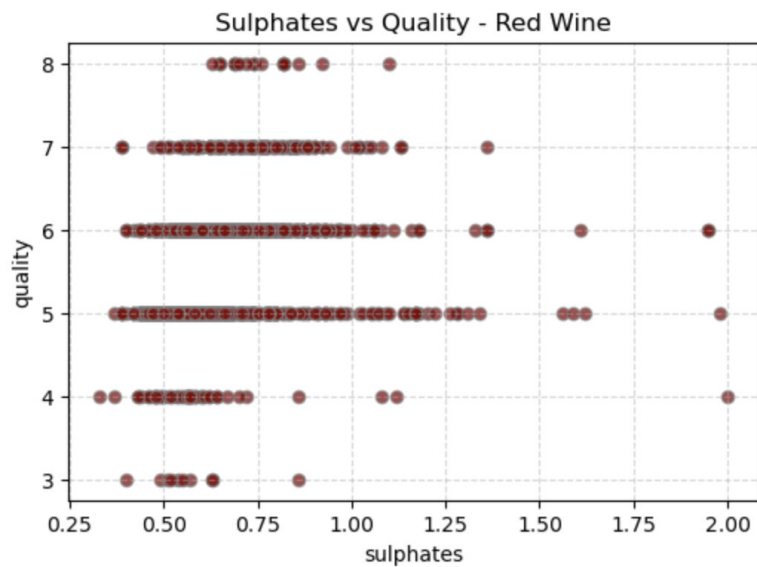


## White Wine

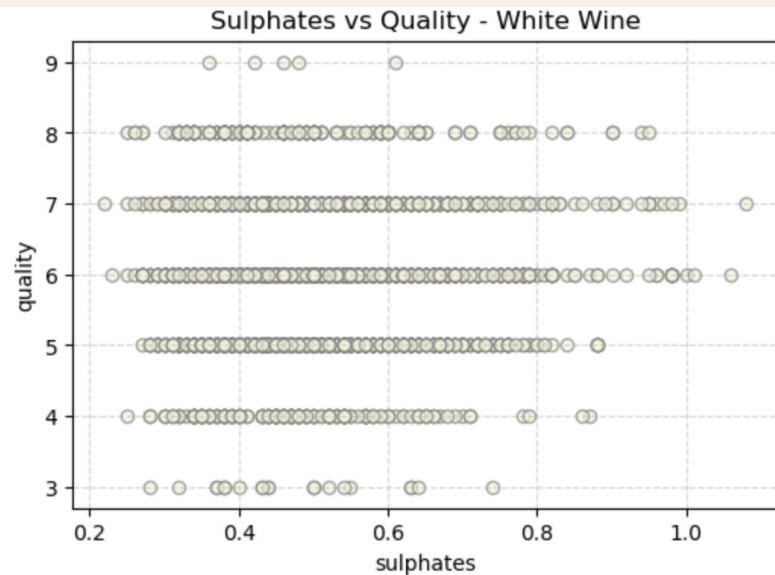


# EDA: Sulphates vs Quality Scatterplot

Red Wine



White Wine



# Key Steps



01

EDA

Conducted EDA to determine our question and that we needed to run a regression



02

Preprocessin  
g

Split 80% train 20% test. Scaled the data since each variable has its own scale



0

3

Hyperparameter  
s

Selected hyperparameters to use to reduce overfitting



04

Train

Trained the models using GridSearchCV with MSE as the scoring method and 10-fold cross-validation

# Hyperparameters



*1st*

**GradientBoosingRegress**

**Hyperparameters:**

n\_estimators, max\_depth,  
learning\_rate, subsample

*2nd*

**RandomForestRegress**

**Hyperparameters:**

n\_estimators, max\_depth,  
max\_features,  
min\_samples\_split,  
min\_samples\_leaf

*3rd*

**BaggingRegressor**

**Hyperparameters:**

n\_estimators, max\_samples,  
max\_features

*d*

# Model Training

*Model Training*



# Red Wine

*Red Wine*



# GradientBoostingRegressor

## Parameter grid:

- 'n\_estimators': [100, 200, 300]
- 'max\_depth': [2, 3]
- 'learning\_rate': [0.01, 0.05]
- 'subsample': [0.7, 0.8, 1.0]

## Best hyperparameters:

- 'n\_estimators': 300
- 'max\_depth': 3
- 'learning\_rate': 0.05
- 'subsample': 0.8

## Best MSE:

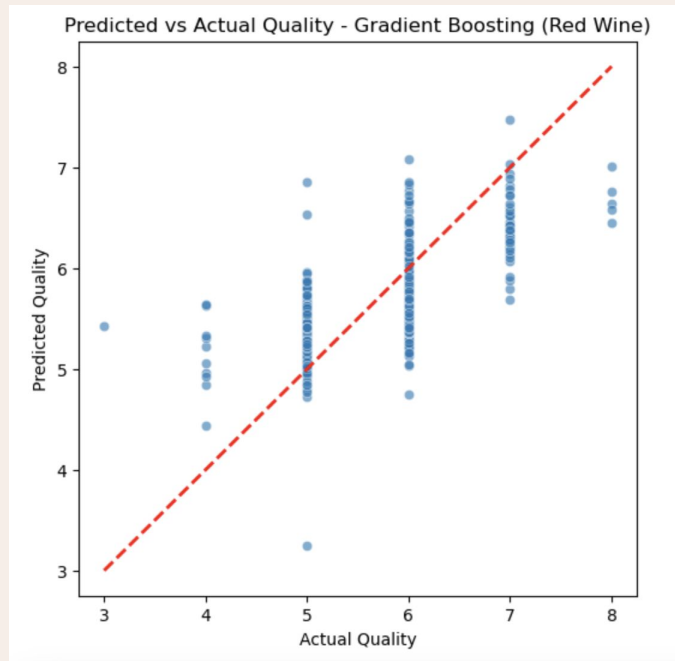
- 0.387



# Model Testing

## Evaluation Metrics:

- R2 on training set: 0.707
- R2 on test set: 0.466
- MSE on test set: 0.349
- RMSE on test set: 0.591
- MAE on test set: 0.469



# RandomForestRegressor

With bootstrapping

## Parameter grid:

- 'n\_estimators': [200, 400]
- 'max\_depth': [3, 5, 8]
- 'max\_features': ['sqrt', 'log2']
- 'min\_samples\_split': [5, 10]
- 'min\_samples\_leaf': [2, 4]

## Best hyperparameters:

- 'n\_estimators': 400
- 'max\_depth': 8
- 'max\_features': 'sqrt'
- 'min\_samples\_split': 5
- 'min\_samples\_leaf': 2

## Best MSE:

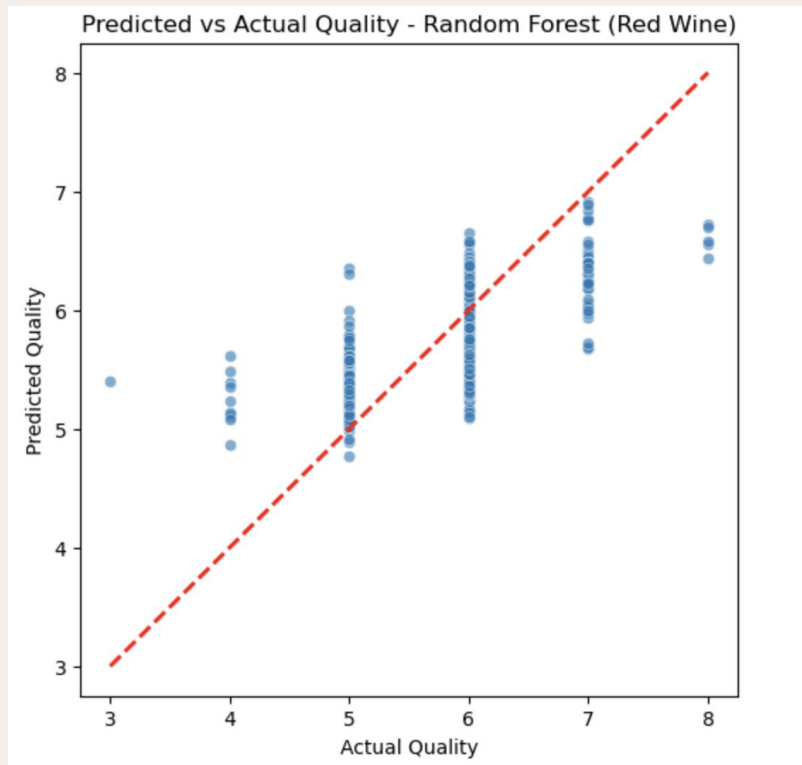
- 0.380



# Model Testing

## Evaluation Metrics:

- $R^2$  on training set: 0.687
- $R^2$  on test set: 0.490 (**highest**)
- MSE on test set: 0.333 (**lowest**)
- RMSE on test set: 0.577
- MAE on test set: 0.465



# BaggingRegressor

**Bagging Model:** DecisionTreeRegressor

- max\_depth=4
- min\_samples\_split=10
- min\_samples\_leaf=4
- random\_state=42

**Parameter grid:**

- 'n\_estimators': [100, 300]
- 'max\_samples': [0.6, 0.8]
- 'max\_features': [0.7, 1.0]

**Best hyperparameters:**

- 'n\_estimators': 300
- 'max\_samples': 0.6
- 'max\_features': 1.0

**Best MSE:**

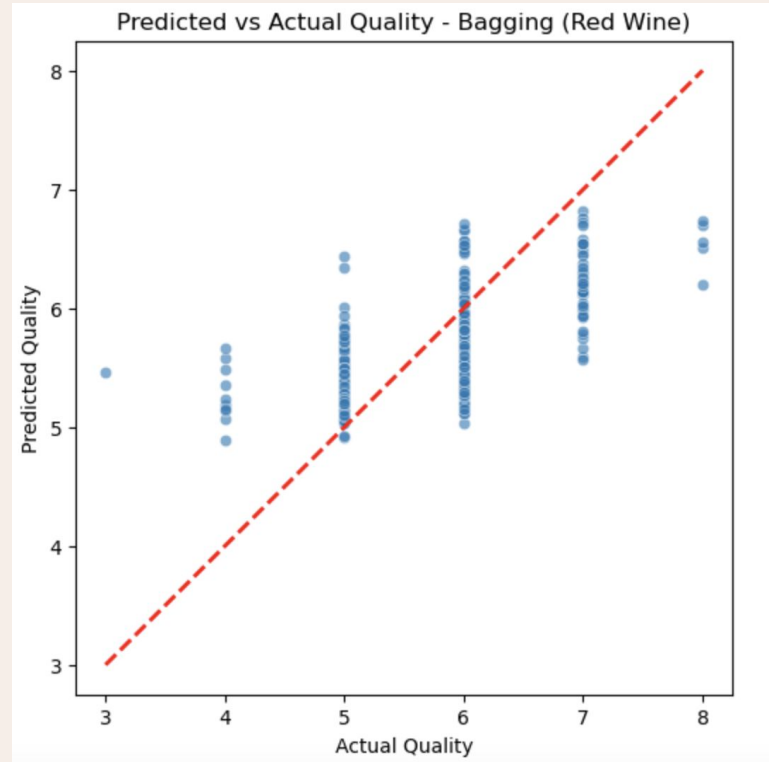
- 0.408



# Model Testing

## Evaluation Metrics:

- $R^2$  on training set: 0.465
- $R^2$  on test set: 0.417
- MSE on test set: 0.381
- RMSE on test set: 0.617
- MAE on test set: 0.500



# White Wine

*White Wine*



# GradientBoostingRegressor

## Parameter grid:

- 'n\_estimators': [100, 200, 300]
- 'max\_depth': [2, 3]
- 'learning\_rate': [0.01, 0.05]
- 'subsample': [0.7, 0.8, 1.0]

## Best hyperparameters:

- 'n\_estimators': 300
- 'max\_depth': 3
- 'learning\_rate': 0.05
- 'subsample': 0.7

## Best MSE:

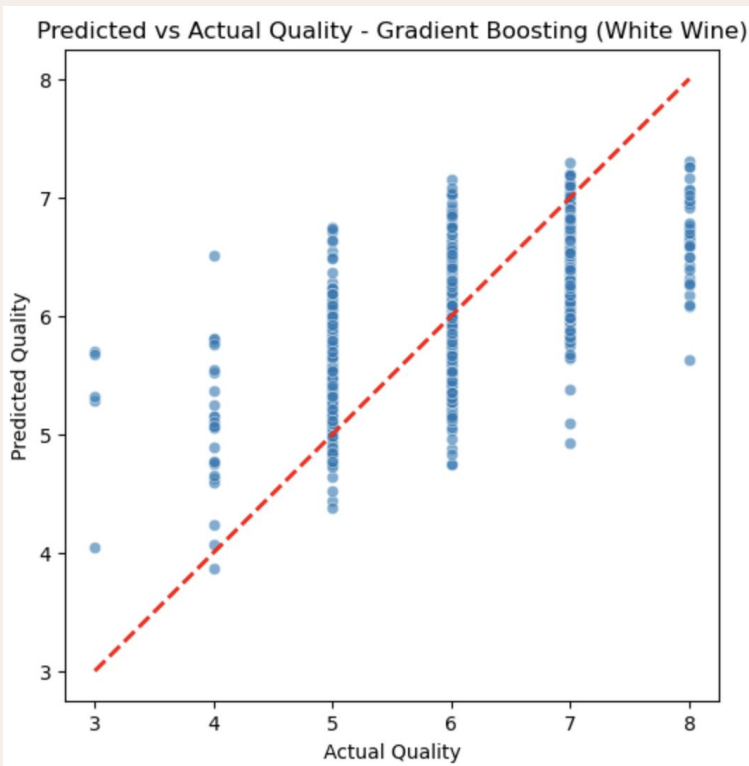
- 0.473



# Model Testing

## Evaluation Metrics:

- $R^2$  on training set: 0.544
- $R^2$  on test set: 0.426
- MSE on test set: 0.444 (**lowest**)
- RMSE on test set: 0.667
- MAE on test set: 0.522



# RandomForestRegressor

With bootstrapping

## Parameter grid:

- 'n\_estimators': [200, 400]
- 'max\_depth': [3, 5, 8]
- 'max\_features': ['sqrt', 'log2']
- 'min\_samples\_split': [5, 10]
- 'min\_samples\_leaf': [2, 4]

## Best hyperparameters:

- 'n\_estimators': 200
- 'max\_depth': 8
- 'max\_features': 'sqrt'
- 'min\_samples\_split': 5
- 'min\_samples\_leaf': 2

## Best MSE:

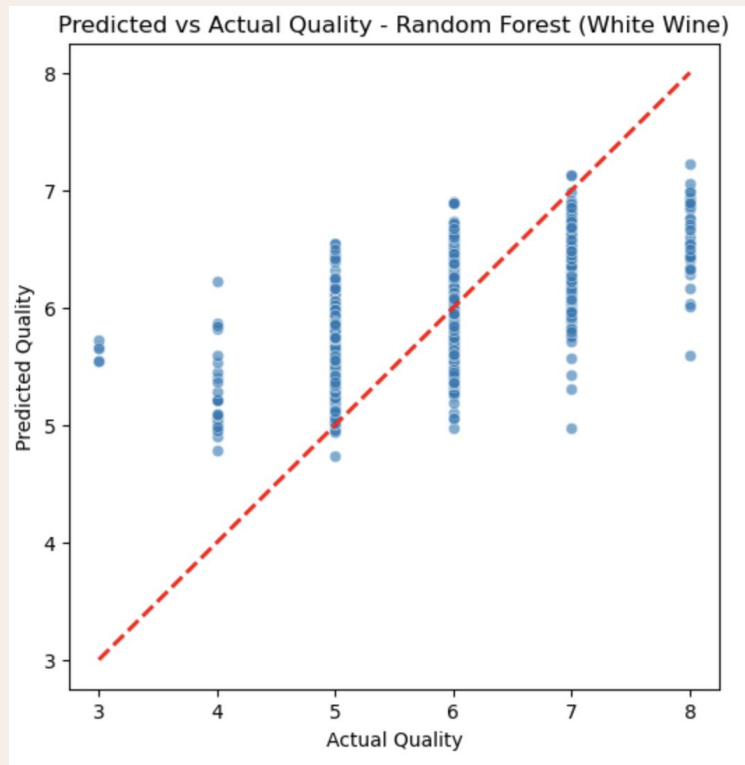
- 0.469



# Model Testing

## Evaluation Metrics:

- $R^2$  on training set: 0.559
- $R^2$  on test set: 0.424 (**highest**)
- MSE on test set: 0.446
- RMSE on test set: 0.668
- MAE on test set: 0.526



# BaggingRegressor

**Bagging Model:** DecisionTreeRegressor

- max\_depth=4
- min\_samples\_split=10
- min\_samples\_leaf=4
- random\_state=42

**Parameter grid:**

- 'n\_estimators': [100, 300]
- 'max\_samples': [0.6, 0.8]
- 'max\_features': [0.7, 1.0]

**Best hyperparameters:**

- 'n\_estimators': 300
- 'max\_samples': 0.6
- 'max\_features': 1.0

**Best MSE:**

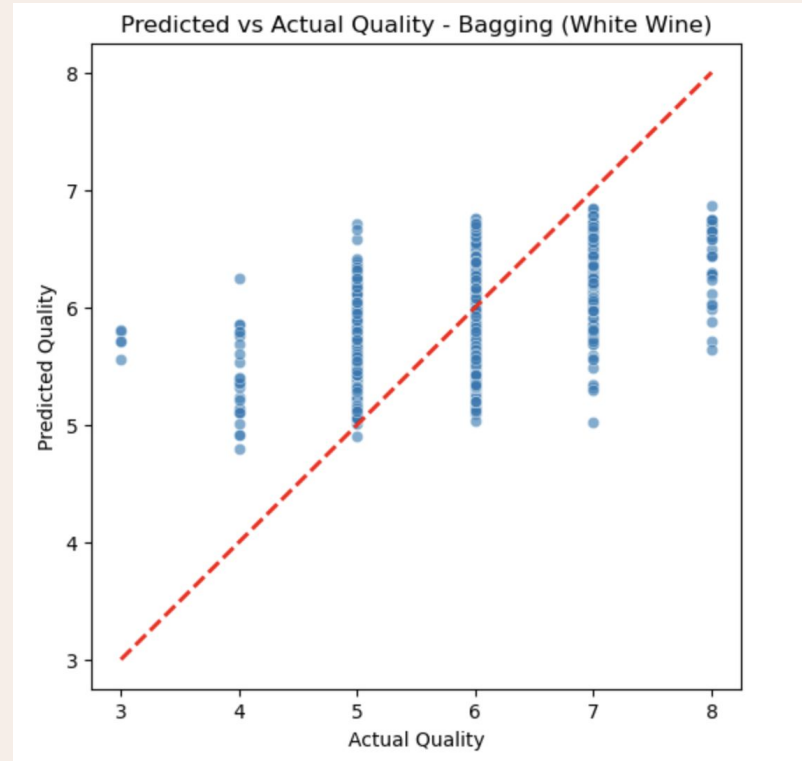
- 0.531



# Model Testing

## Evaluation Metrics:

- $R^2$  on training set: 0.364
- $R^2$  on test set: 0.331
- MSE on test set: 0.518
- RMSE on test set: 0.720
- MAE on test set: 0.572



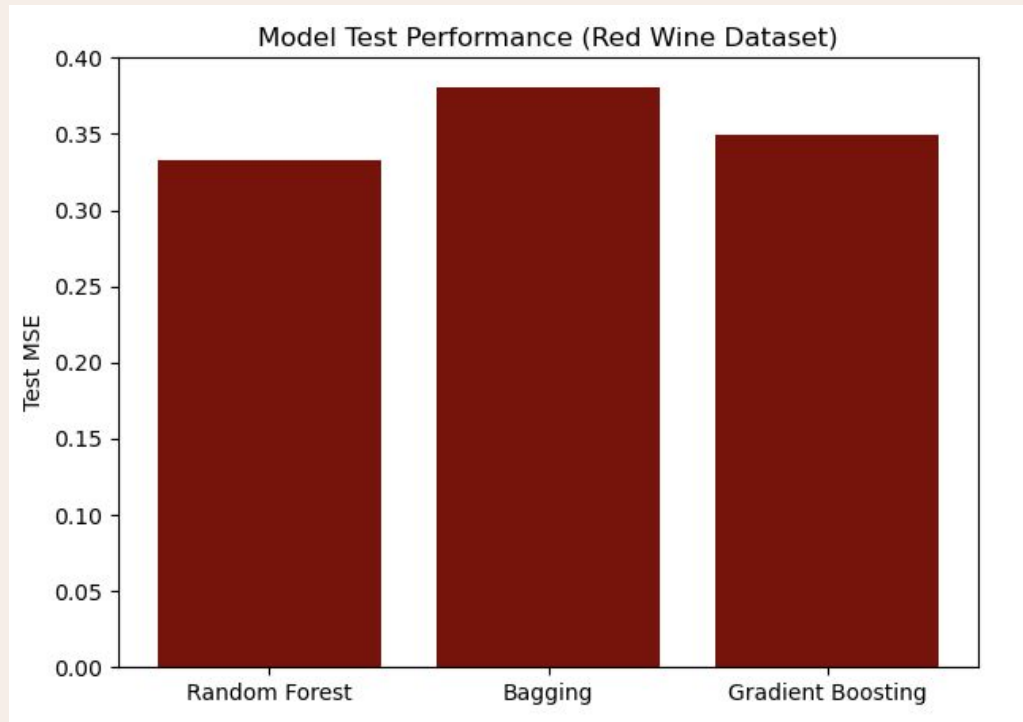
# Model Comparison n

*Model Comparison*



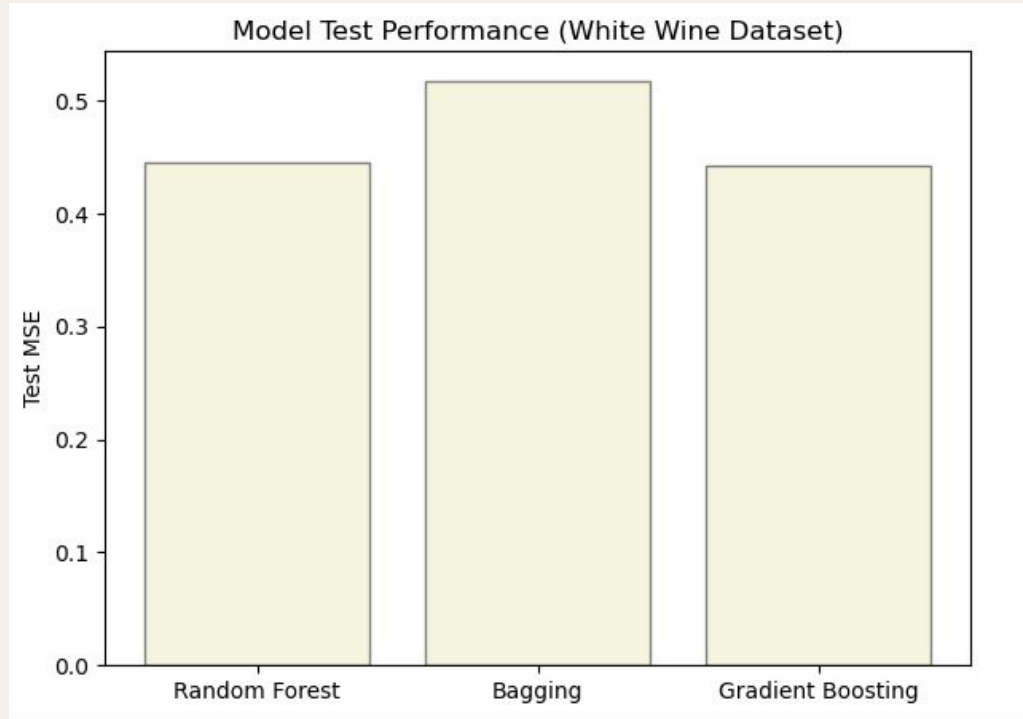
# Model Performance Comparison (Red Wine)

- Random Forest achieved the highest R-squared and the lowest MSE
- Gradient Boosting performed similarly but was slightly worse
- Bagging had the lowest predictive performance



# Model Performance Comparison (White Wine)

- Random Forest again showed strong performance
- Gradient Boosting had similar accuracy (0.004 better than Random Forest)
- Bagging performed the worst across metrics



# Overall Model Comparison

- Random Forest produced the lowest prediction error across both datasets
- Ensemble tree models captured non-linear relationships between chemical attributes and quality
- All three models performed similarly, but Random Forest consistently had the best generalization



# Results Comparison n

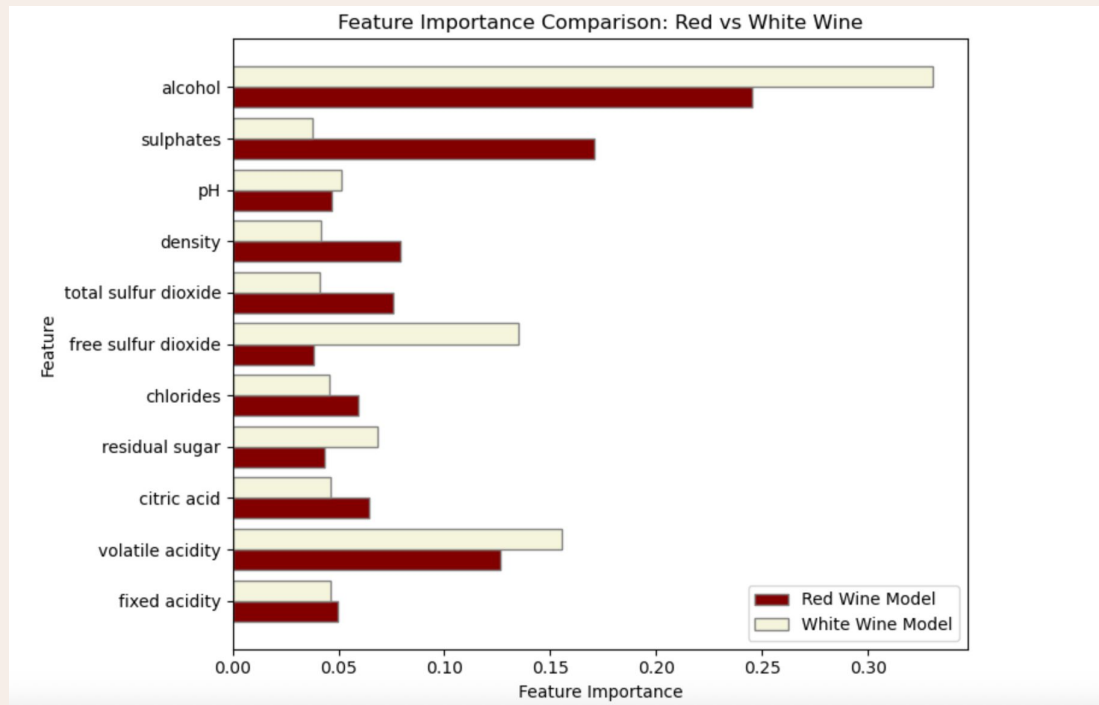
*Results Comparison*



# Comparison of Datasets

## Takeaways:

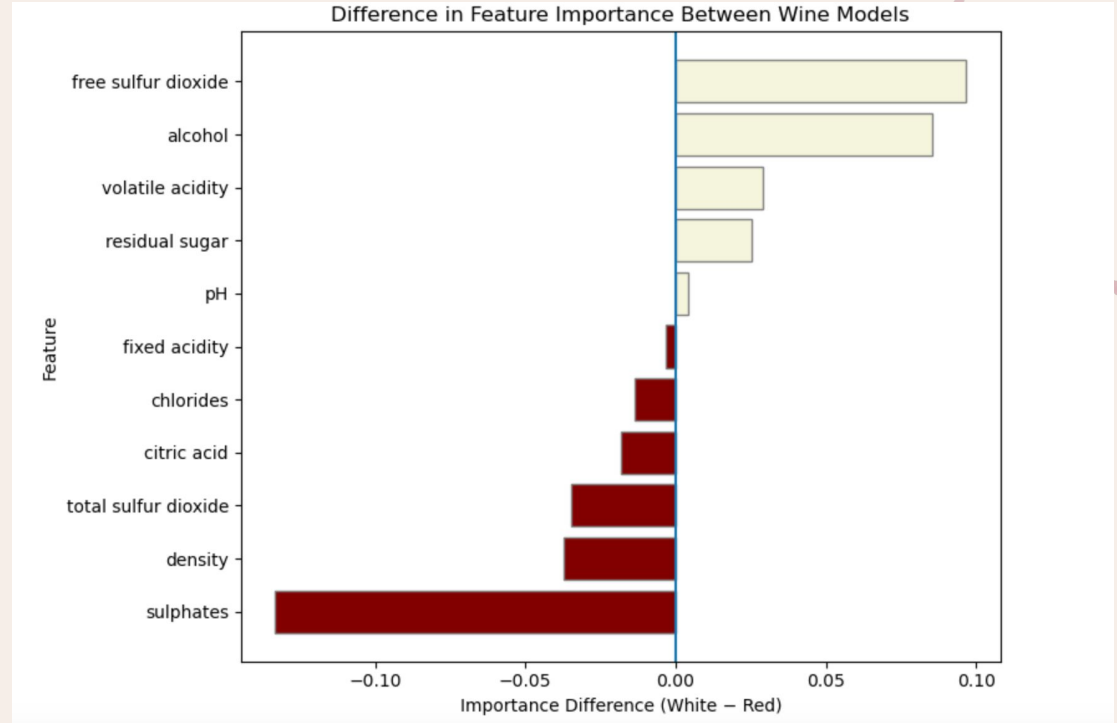
- Alcohol most important for both
- Sulfates affect **red wine**
- Sulfur affects **white wine**
- Acidity affects the quality of both wines similarly



# Comparison of Datasets

## Takeaways:

- Acids have less of a difference between wine types
- **White:** sulfur, alcohol
- **Red:** sulphates influence
- Wine differences shown



# Recommendations - Wine Producers



## Sulphate Optimization

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Sulphate is a very important feature for red wine quality, so avoid under-dosing preservatives



## Alcohol Level Control

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Alcohol is correlated with an increase in quality for both wines, so focus on grape maturity



## Balance Acidity

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The various acidities are important features are important for both wines, so improving fermentation control and minimizing bacterial contamination is crucial

*Recommend*



# Recommendations - Wine Sellers

## Market Wines using Specific Wording

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Product Positioning and  
highlighting certain features can  
help increase wine sales

### **“Full-bodied”**

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A full-bodied wine refers to  
one with a high alcohol  
content

### **“Structured”**

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Balanced amount of acid  
and alcohol

### **“Bright Acidity”**

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Both wines had various  
types of acidity as  
important features



# Conclusion

- Random Forest produced most accurate prediction and had the best fit for both datasets—can be used to improve wine evaluation and production decisions.
- Wine Sellers should highlight most important features of wines when marketing
- Wine Producers should focus on steps to perfect production of wines with a good balance of their most important features



Thank  
you!

